Sheets

RP2040
File: rp2040.kicad_sch

FPGA
File: fpga.kicad_sch

FPGA2
File: fpga2.kicad_sch

VGA
File: vga.kicad_sch

FDC
File: fdc.kicad_sch

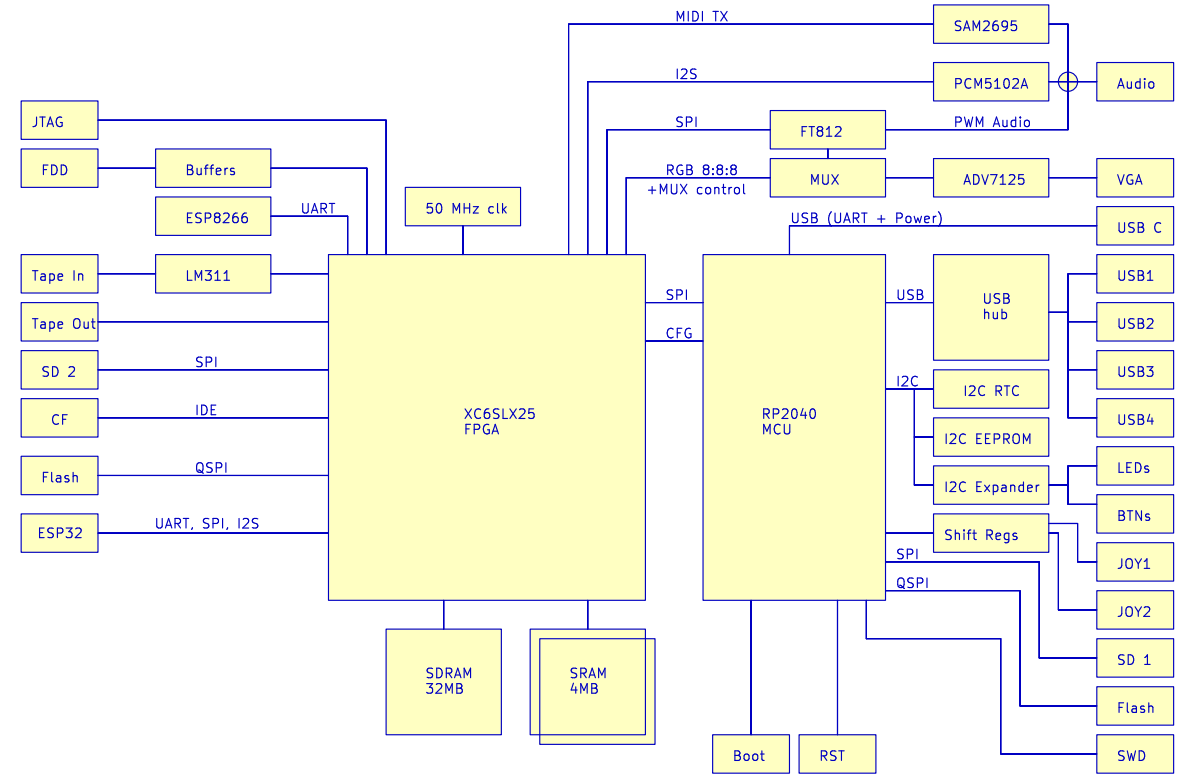
Storage
File: storage.kicad_sch

Interfaces
File: interfaces.kicad_sch

Audio
File: audio.kicad_sch

ESP32
File: esp32.kicad_sch

Functional schematics

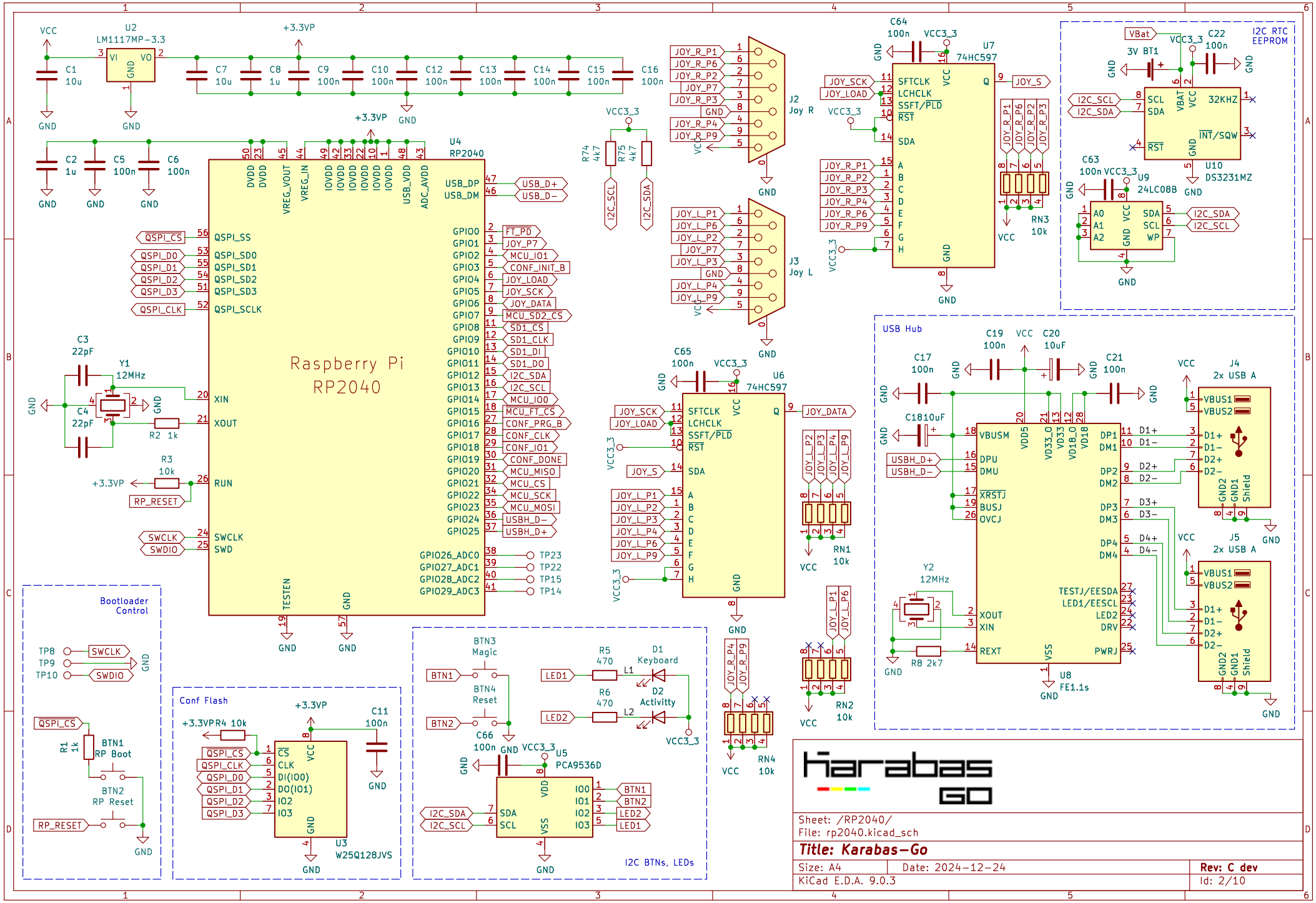


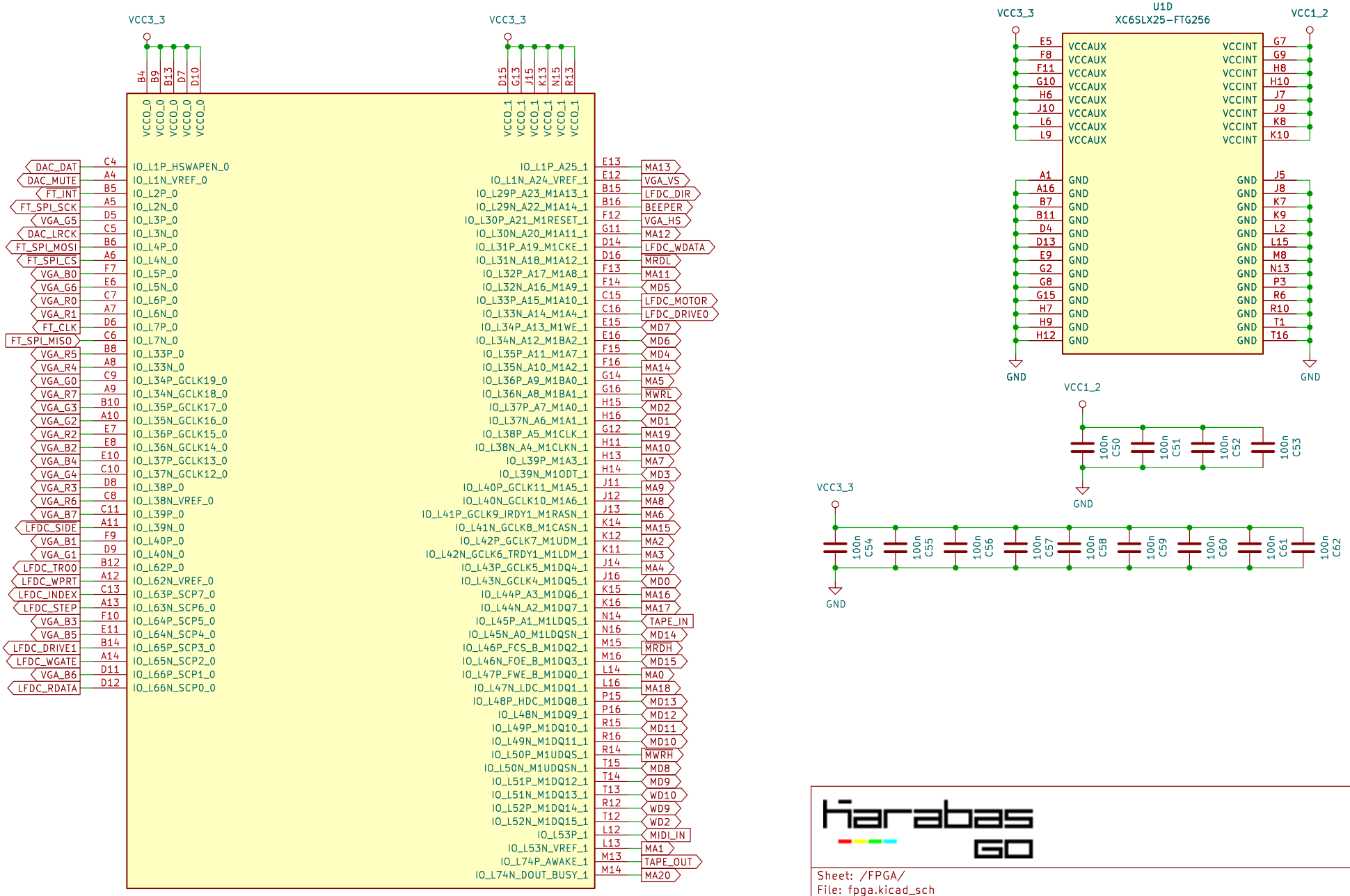
H1 MountingHole — GND
H3 MountingHole — GND
H5 MountingHole — GND
H7 MountingHole — GND
H2 MountingHole — GND
H4 MountingHole — GND
H6 MountingHole — GND
H8 MountingHole — GND

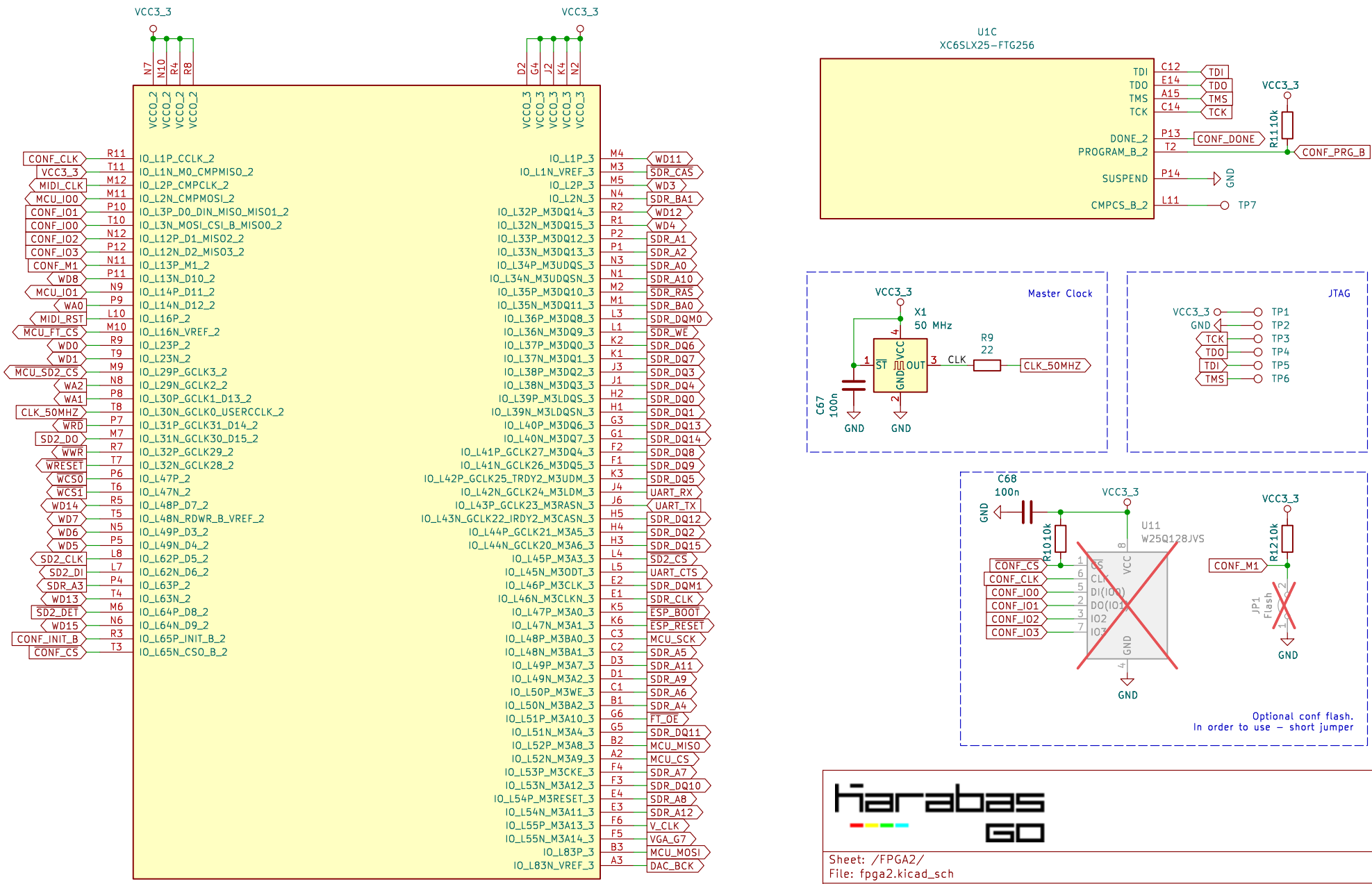
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Title: Karabas-Go

Size: A4	Date: 2024-12-24	**Rev: C dev**
KiCad E.D.A. 9.0.3		Id: 1/10


Raspberry Pi
RP2040
U4
RP2040
U2
LM1117MP-3.3
U7
74HC597
U6
74HC597
U5
PCA9536D
U3
W25Q128JVS
U8
FE1.1s
U9
24LC08B
U10
DS3231MZ
J2
Joy R
J3
Joy L
J4
2x USB A
J5
2x USB A
Y1
12MHz
Y2
12MHz
USB Hub
I2C RTC EEPROM
Bootloader Control
Conf Flash
I2C BTNs, LEDs
BTN3
Magic
BTN4
Reset
BTN1
RP Boot
BTN2
RP Reset
D1
Keyboard
D2
Activitty
Sheet: /RP2040/
File: rp2040.kicad_sch
Title: Karabas-Go
Size: A4
Date: 2024-12-24
Rev: C dev
KiCad E.D.A. 9.0.3
Id: 2/10

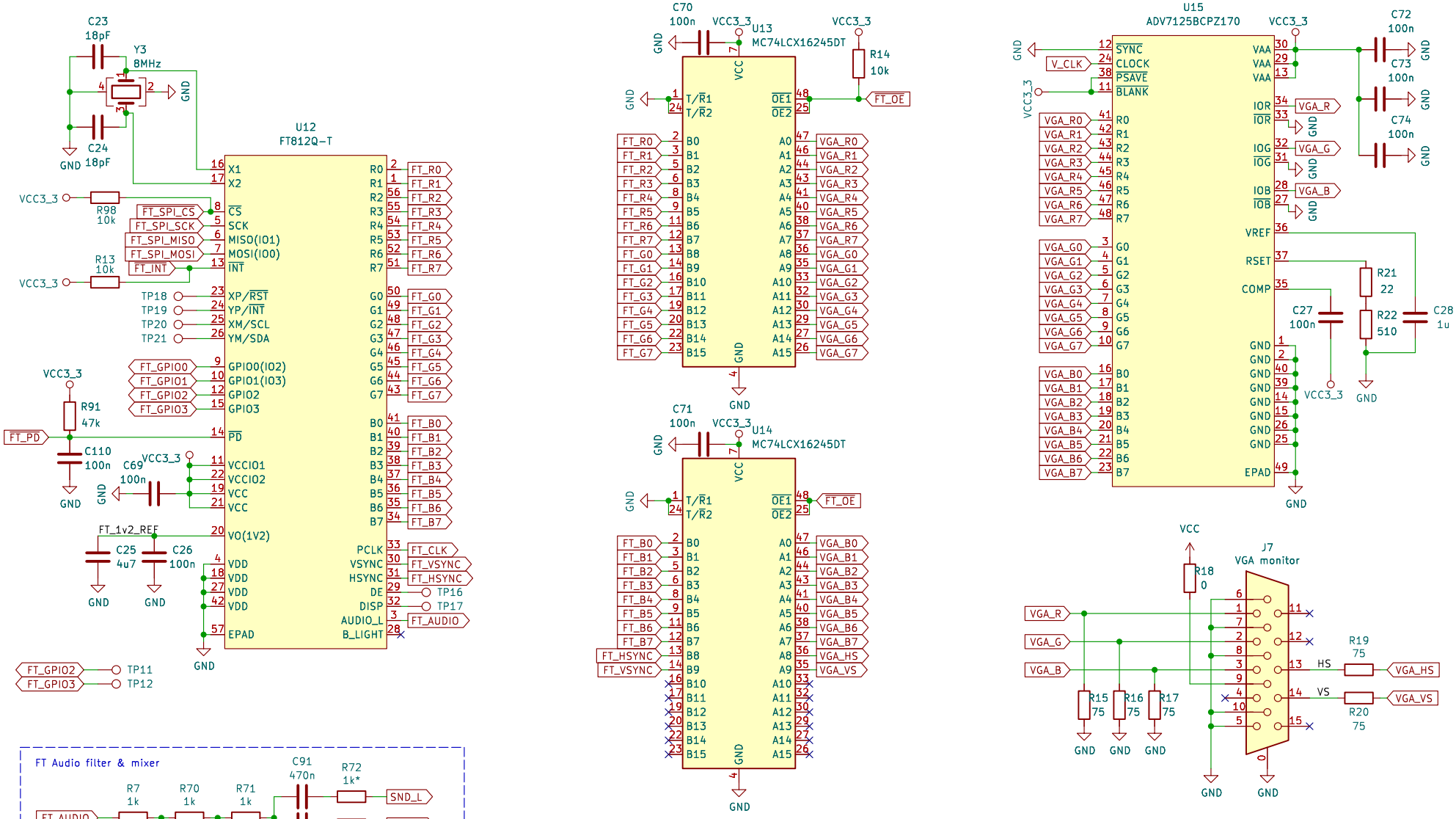

VCC3_3
B4
B9
B13
D7
D10
VCCO_0
VCC3_3
D15
G13
J15
K13
N15
R13
VCCO_1
C4 IO_L1P_HSWAPEN_0
A4 IO_L1N_VREF_0
B5 IO_L2P_0
A5 IO_L2N_0
D5 IO_L3P_0
C5 IO_L3N_0
B6 IO_L4P_0
A6 IO_L4N_0
F7 IO_L5P_0
E6 IO_L5N_0
C7 IO_L6P_0
A7 IO_L6N_0
D6 IO_L7P_0
C6 IO_L7N_0
B8 IO_L33P_0
A8 IO_L33N_0
C9 IO_L34P_GCLK19_0
A9 IO_L34N_GCLK18_0
B10 IO_L35P_GCLK17_0
A10 IO_L35N_GCLK16_0
E7 IO_L36P_GCLK15_0
E8 IO_L36N_GCLK14_0
E10 IO_L37P_GCLK13_0
C10 IO_L37N_GCLK12_0
D8 IO_L38P_0
C8 IO_L38N_VREF_0
C11 IO_L39P_0
A11 IO_L39N_0
F9 IO_L40P_0
D9 IO_L40N_0
B12 IO_L62P_0
A12 IO_L62N_VREF_0
C13 IO_L63P_SCP7_0
A13 IO_L63N_SCP6_0
F10 IO_L64P_SCP5_0
E11 IO_L64N_SCP4_0
B14 IO_L65P_SCP3_0
A14 IO_L65N_SCP2_0
D11 IO_L66P_SCP1_0
D12 IO_L66N_SCP0_0
DAC_DAT
DAC_MUTE
FT_INT
FT_SPI_SCK
VGA_G5
DAC_LRCK
FT_SPI_MOSI
FT_SPI_CS
VGA_B0
VGA_G6
VGA_R0
VGA_R1
FT_CLK
FT_SPI_MISO
VGA_R5
VGA_R4
VGA_G0
VGA_R7
VGA_G3
VGA_G2
VGA_R2
VGA_B2
VGA_B4
VGA_G4
VGA_R3
VGA_R6
VGA_B7
LFDC_SIDE
VGA_B1
VGA_G1
LFDC_TR00
LFDC_WPRT
LFDC_INDEX
LFDC_STEP
VGA_B3
VGA_B5
LFDC_DRIVE1
LFDC_WGATE
VGA_B6
LFDC_RDATA
E13 IO_L1P_A25_1
E12 IO_L1N_A24_VREF_1
B15 IO_L29P_A23_M1A13_1
B16 IO_L29N_A22_M1A14_1
F12 IO_L30P_A21_M1RESET_1
G11 IO_L30N_A20_M1A11_1
D14 IO_L31P_A19_M1CKE_1
D16 IO_L31N_A18_M1A12_1
F13 IO_L32P_A17_M1A8_1
F14 IO_L32N_A16_M1A9_1
C15 IO_L33P_A15_M1A10_1
C16 IO_L33N_A14_M1A4_1
E15 IO_L34P_A13_M1WE_1
E16 IO_L34N_A12_M1BA2_1
F15 IO_L35P_A11_M1A7_1
F16 IO_L35N_A10_M1A2_1
G14 IO_L36P_A9_M1BA0_1
G16 IO_L36N_A8_M1BA1_1
H15 IO_L37P_A7_M1A0_1
H16 IO_L37N_A6_M1A1_1
G12 IO_L38P_A5_M1CLK_1
H11 IO_L38N_A4_M1CLKN_1
H13 IO_L39P_M1A3_1
H14 IO_L39N_M1ODT_1
J11 IO_L40P_GCLK11_M1A5_1
J12 IO_L40N_GCLK10_M1A6_1
J13 IO_L41P_GCLK9_IRDY1_M1RASN_1
K14 IO_L41N_GCLK8_M1CASN_1
K12 IO_L42P_GCLK7_M1UDM_1
K11 IO_L42N_GCLK6_TRDY1_M1LDM_1
J14 IO_L43P_GCLK5_M1DQ4_1
J16 IO_L43N_GCLK4_M1DQ5_1
K15 IO_L44P_A3_M1DQ6_1
K16 IO_L44N_A2_M1DQ7_1
N14 IO_L45P_A1_M1LDQS_1
N16 IO_L45N_A0_M1LDQSN_1
M15 IO_L46P_FCS_B_M1DQ2_1
M16 IO_L46N_FOE_B_M1DQ3_1
L14 IO_L47P_FWE_B_M1DQ0_1
L16 IO_L47N_LDC_M1DQ1_1
P15 IO_L48P_HDC_M1DQ8_1
P16 IO_L48N_M1DQ9_1
R15 IO_L49P_M1DQ10_1
R16 IO_L49N_M1DQ11_1
R14 IO_L50P_M1UDQS_1
T15 IO_L50N_M1UDQSN_1
T14 IO_L51P_M1DQ12_1
T13 IO_L51N_M1DQ13_1
R12 IO_L52P_M1DQ14_1
T12 IO_L52N_M1DQ15_1
L12 IO_L53P_1
L13 IO_L53N_VREF_1
M13 IO_L74P_AWAKE_1
M14 IO_L74N_DOUT_BUSY_1
MA13
VGA_VS
LFDC_DIR
BEEPER
VGA_HS
MA12
LFDC_WDATA
MRDL
MA11
MD5
LFDC_MOTOR
LFDC_DRIVE0
MD7
MD6
MD4
MA14
MA5
MWRL
MD2
MD1
MA19
MA10
MA7
MD3
MA9
MA8
MA6
MA15
MA2
MA3
MA4
MD0
MA16
MA17
TAPE_IN
MD14
MRDH
MD15
MA0
MA18
MD13
MD12
MD11
MD10
MWRH
MD8
MD9
WD10
WD9
WD2
MIDI_IN
MA1
TAPE_OUT
MA20
U1A
XC6SLX25-FTG256
U1D
XC6SLX25-FTG256
VCC3_3
VCC1_2
E5 VCCAUX
F8 VCCAUX
F11 VCCAUX
G10 VCCAUX
H6 VCCAUX
J10 VCCAUX
L6 VCCAUX
L9 VCCAUX
G7 VCCINT
G9 VCCINT
H8 VCCINT
H10 VCCINT
J7 VCCINT
J9 VCCINT
K8 VCCINT
K10 VCCINT
A1 GND
A16 GND
B7 GND
B11 GND
D4 GND
D13 GND
E9 GND
G2 GND
G8 GND
G15 GND
H7 GND
H9 GND
H12 GND
J5 GND
J8 GND
K7 GND
K9 GND
L2 GND
L15 GND
M8 GND
N13 GND
P3 GND
R6 GND
R10 GND
T1 GND
T16 GND
GND
VCC1_2
100n C50
100n C51
100n C52
100n C53
VCC3_3
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100n C55
100n C56
100n C57
100n C58
100n C59
100n C60
100n C61
100n C62
GND
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Size: A4
Date: 2024-12-24
Rev: C dev
KiCad E.D.A. 9.0.3
Id: 3/10

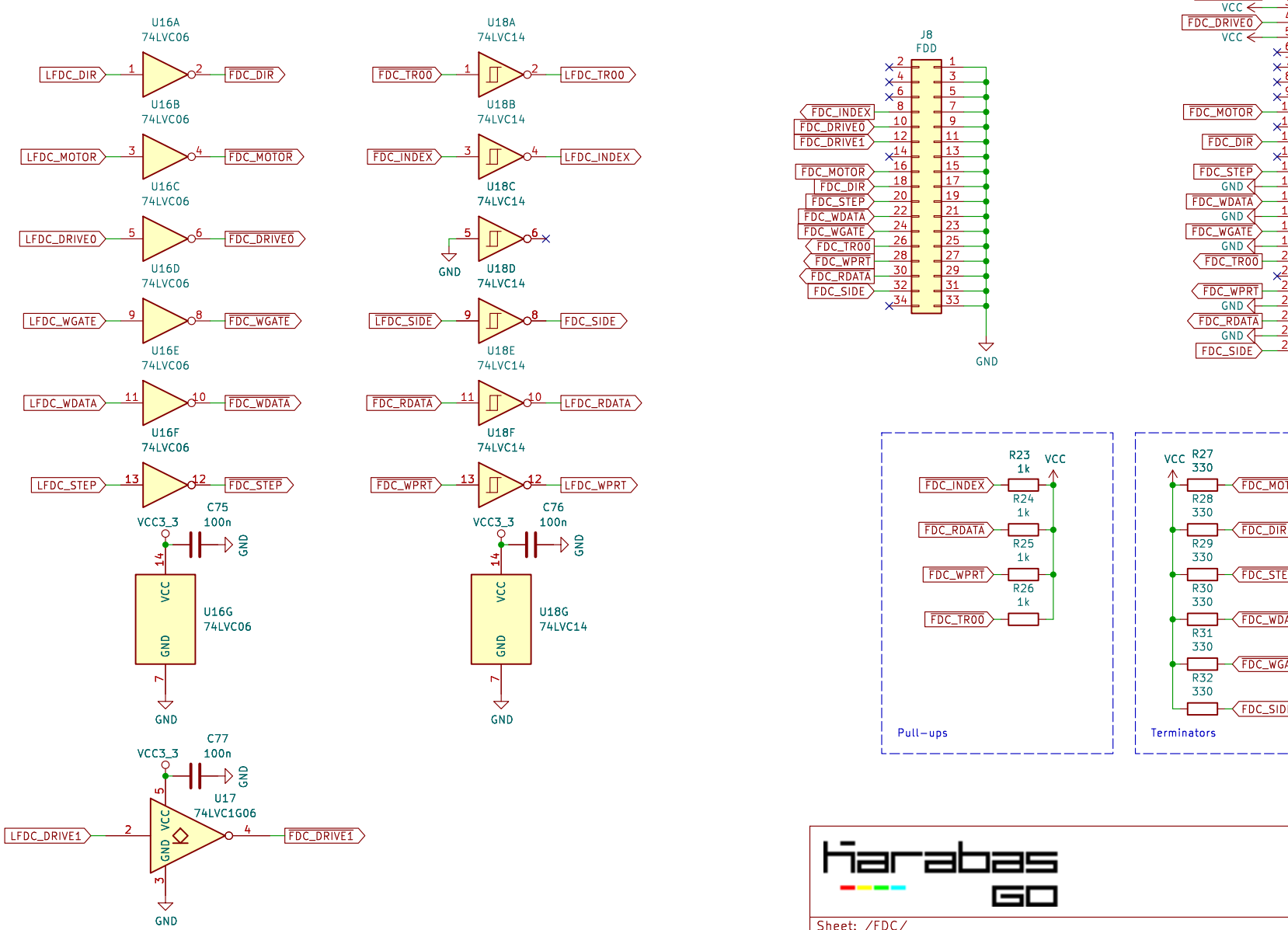


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KiCad E.D.A. 9.0.3		Id: 4/10

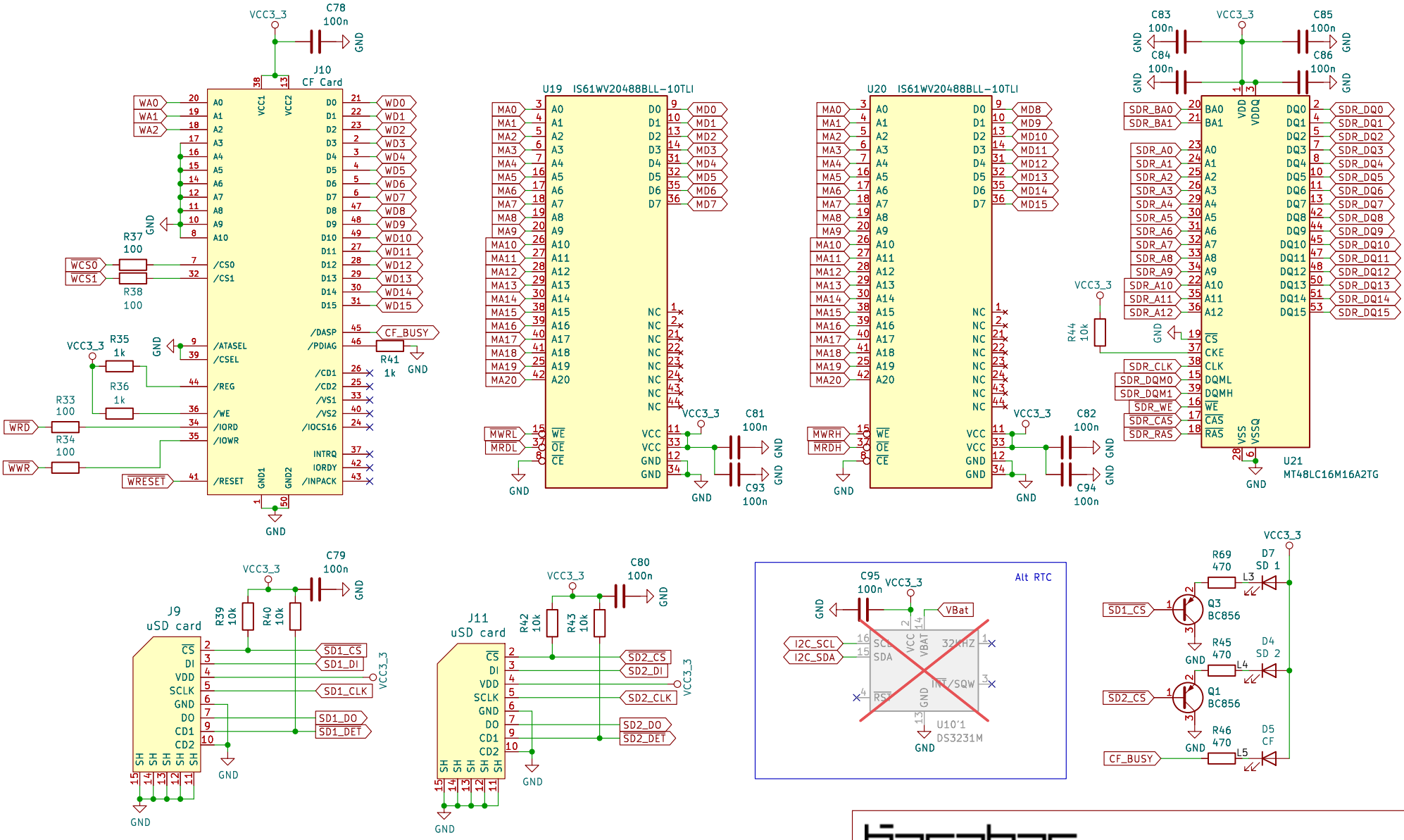

C23
18pF
Y3
8MHz
C24
18pF
GND
U12
FT812Q-T
X1
X2
CS
SCK
MISO(IO1)
MOSI(IO0)
INT
XP/RST
YP/INT
XM/SCL
YM/SDA
GPIO0(IO2)
GPIO1(IO3)
GPIO2
GPIO3
PD
VCCIO1
VCCIO2
VCC
VCC
VO(1V2)
VDD
VDD
VDD
VDD
EPAD
R0
R1
R2
R3
R4
R5
R6
R7
G0
G1
G2
G3
G4
G5
G6
G7
B0
B1
B2
B3
B4
B5
B6
B7
PCLK
VSYNC
HSYNC
DE
DISP
AUDIO_L
B_LIGHT
VCC3_3
R98
10k
R13
10k
FT_SPI_CS
FT_SPI_SCK
FT_SPI_MISO
FT_SPI_MOSI
FT_INT
TP18
TP19
TP20
TP21
FT_GPIO0
FT_GPIO1
FT_GPIO2
FT_GPIO3
R91
47k
FT_PD
C110
100n
C69
100n
FT_1v2_REF
C25
4u7
C26
100n
TP11
TP12
FT_CLK
FT_VSYNC
FT_HSYNC
TP16
TP17
FT_AUDIO
FT Audio filter & mixer
R7
1k
R70
1k
R71
1k
C91
470n
R72
1k*
R73
1k*
C88
4n7
C89
4n7
C90
4n7
C92
470n
SND_L
SND_R
C70
100n
U13
MC74LCX16245DT
R14
10k
FT_OE
C71
100n
U14
MC74LCX16245DT
U15
ADV7125BCPZ170
SYNC
CLOCK
PSAVE
BLANK
V_CLK
VAA
IOR
IOG
IOB
VREF
RSET
COMP
EPAD
C72
100n
C73
100n
C74
100n
VGA_R
VGA_G
VGA_B
R21
22
R22
510
C27
100n
C28
1u
J7
VGA monitor
R18
0
R19
75
R20
75
R15
75
R16
75
R17
75
HS
VS
VGA_HS
VGA_VS
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Title: Karabas-Go
Size: A4
Date: 2024-12-24
Rev: C dev
KiCad E.D.A. 9.0.3
Id: 5/10



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Title: Karabas-Go

Size: A4	Date: 2024-12-24	**Rev: C dev**
KiCad E.D.A. 9.0.3		Id: 6/10

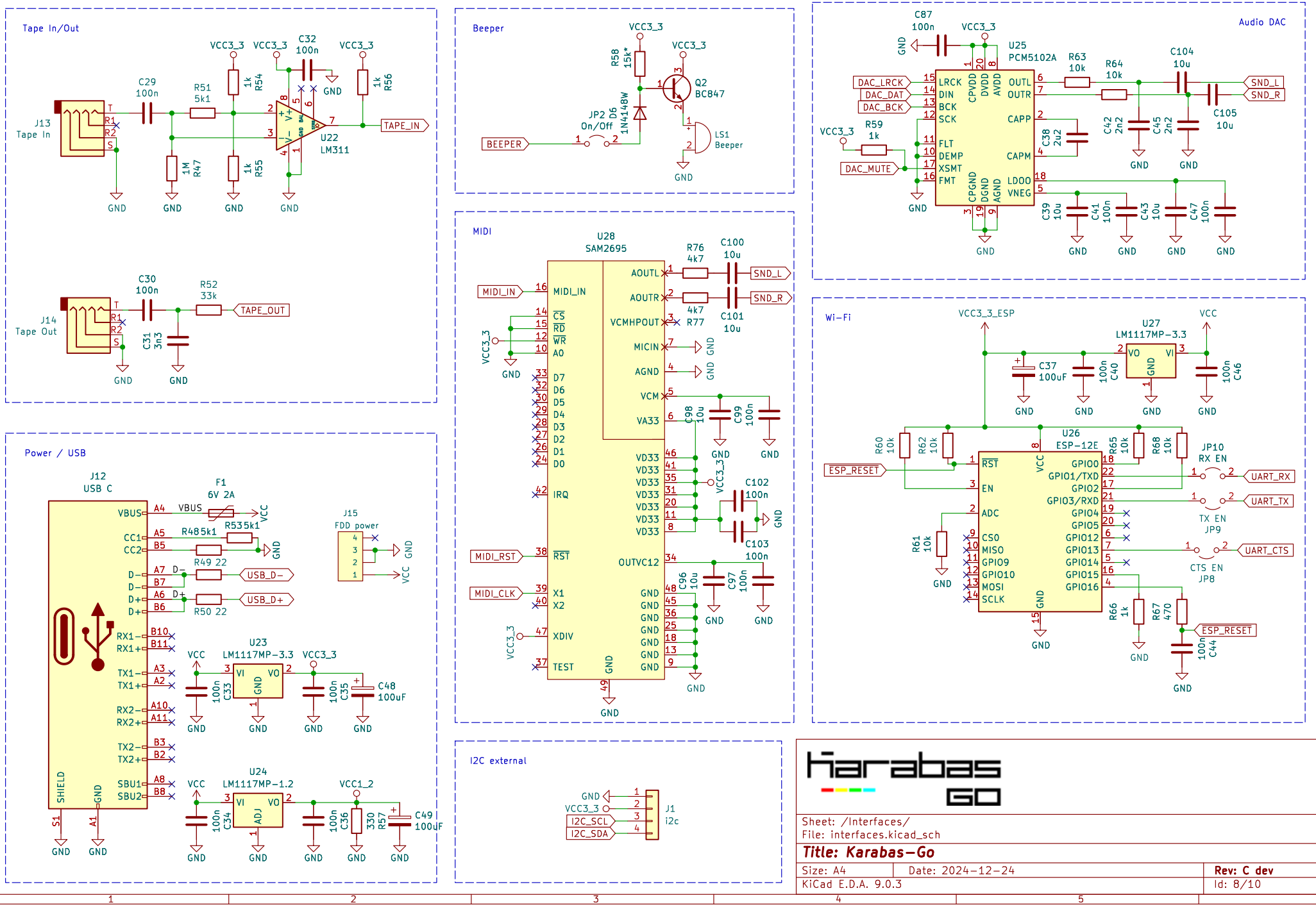


Harabas GO

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KiCad E.D.A. 9.0.3		Id: 7/10

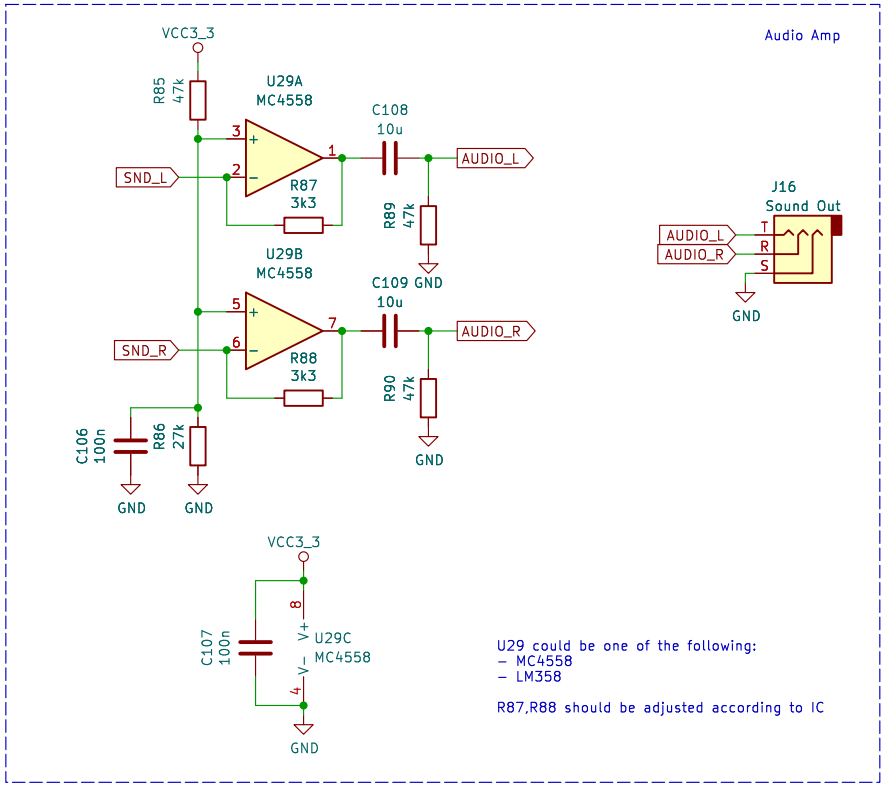


Harabas GO

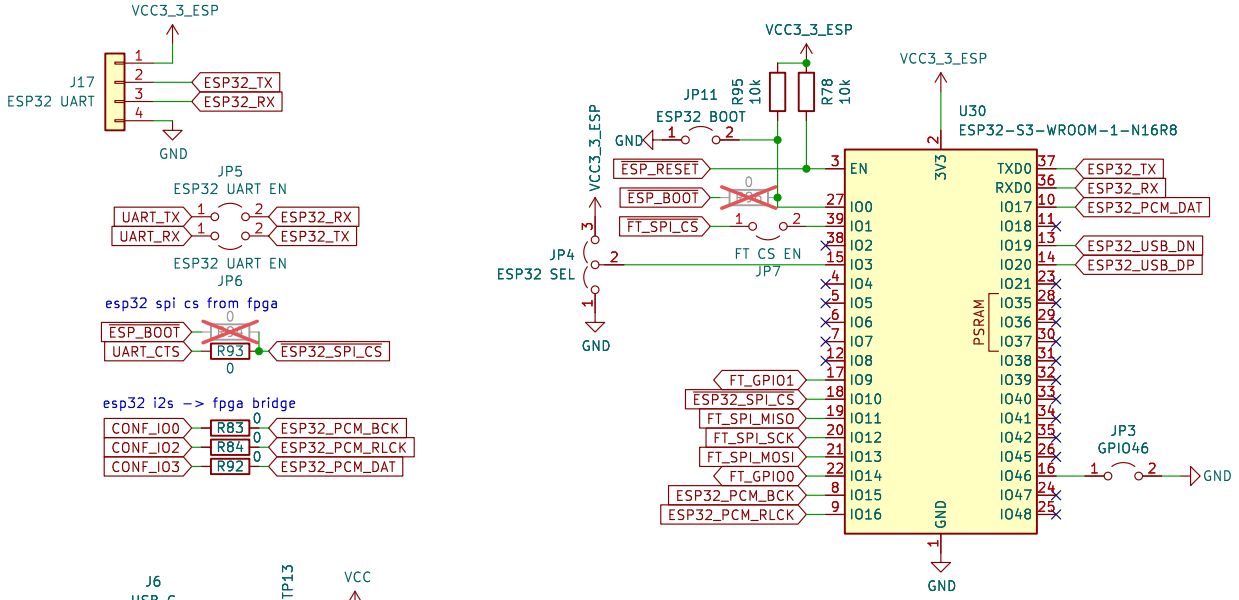
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KiCad E.D.A. 9.0.3		Id: 9/10



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